

LESSON 6-1 • SESSION 1 • PRACTICE • FORM B

Dividing by a Unit Fraction

Learning target – I can divide a whole number by a unit fraction and explain what the quotient counts. • 6.NS.A.1

REMEMBER

- A **unit fraction** has 1 on top, like $1/3$. Dividing by it asks: **how many pieces of that size fit?**
- **Keep, Change, Flip:** keep the first number, change $\div$ to $\times$, flip the divisor. $4 \div 1/3 = 4 \times 3$.
- The quotient is **greater** than the number you start with. Check: quotient $\times$ divisor = start.

WORKED EXAMPLE

A cook has 4 cups of broth. How many $1/3$ -cup servings can she pour?

Step 1 Write the division: $4 \div 1/3$.

Step 2 Keep 4, change $\div$ to $\times$, flip $1/3$ to 3: $4 \times 3 = 12$.

Step 3 Check: $12 \times 1/3 = 4$ cups.

Answer 12 servings

WORDS unit fraction (*fracción unitaria*) • reciprocal (*recíproco*) • quotient (*cociente*)

PRACTICE

show your work • label units

1. Divide. Think: how many of the unit fraction fit into the whole number?

a. $7 \div 1/2 =$ _____

b. $2 \div 1/4 =$ _____

c. $8 \div 1/3 =$ _____

d. $4 \div 1/5 =$ _____

e. $6 \div 1/10 =$ _____

f. $5 \div 1/8 =$ _____

2. **Oats and Seeds.** Ms. Patel has 6 pounds of oats and 3 pounds of seeds. Each bar is 1 pound.



1 whole

1 whole

1 whole

- a. Each bird feeder uses $1/3$ pound of oats. How many feeders? _____ feeders

- b. She splits the seeds into $1/4$ -pound packets. Use the bars: how many packets? _____ packets

3. **Garden Edging.** A gardener has 10 feet of edging. Each small bed needs $1/4$ foot. How many beds can he edge?

Hint Each foot holds 4 fourths.

Show your work.

Answer _____ beds

4. **Frozen Yogurt.** A snack cart has 4 quarts of frozen yogurt. Each sample cup holds $1/6$ quart. How many cups can it fill?

Hint Count sixths: 6 in each quart.

Show your work.

Answer _____ cups

5. Water Bottles. A camp has 9 gallons of water. Each bottle holds $\frac{1}{2}$ gallon. Theo says only 4.5 bottles can be filled. Justify or correct his answer.

Hint Ask: how many halves are in 9 wholes?

Show your work.

Answer_____ bottles

6. Tape Pieces. Keisha cuts a 3-yard roll of tape into pieces that are each $\frac{1}{10}$ yard long. She gives 12 pieces to a friend.

Hint Divide first, then subtract the 12 pieces.

- a.** How many pieces does she cut? _____ pieces
- b.** How many are left? _____ pieces

Show your work.

7. Craft Table. A teacher has 10 meters of yarn for bracelets.

- a.** Each bracelet uses $\frac{1}{5}$ meter. How many bracelets? _____ bracelets
- b.** Each bracelet uses $\frac{1}{2}$ meter instead. How many bracelets? _____ bracelets
- c.** Why does a larger piece for each bracelet give fewer bracelets? _____

Show your work.

8. Explain It. Find $7 \div \frac{1}{5}$.

Hint How many fifths are in 1 whole? In 7 wholes?

- a.** Write the multiplication that matches this division and solve. _____
- b.** Explain why the answer is greater than 7. _____

FIND THE ERROR

explain the mistake, then fix it

9. Amir’s Work. Amir solved $4 \div \frac{1}{5}$. He flipped the first number and wrote $\frac{1}{4} \times \frac{1}{5} = \frac{1}{20}$. Explain his mistake. Then find the correct quotient.

TEST PRACTICE

choose the best answer

10. Find $3 \div \frac{1}{6}$.

- A** $\frac{1}{2}$ **B** $\frac{1}{18}$ **C** 9 **D** 18

11. The reciprocal of $\frac{1}{9}$ is –

- A** $\frac{1}{9}$ **B** 9 **C** 0.9 **D** –9

SELF-CHECK CODE

find your answer · copy its letter into that problem’s box · spell a math word

2/3 Z	40 G	4.5 Q	30 U	18 O	50 P	42 X	35 S	24 R
#3		#4		#5		#6a		#7a
								#8a

CHECK YOURSELF ☐ I can do this on my own ☐ I need a little help ☐ I have a question about #_____